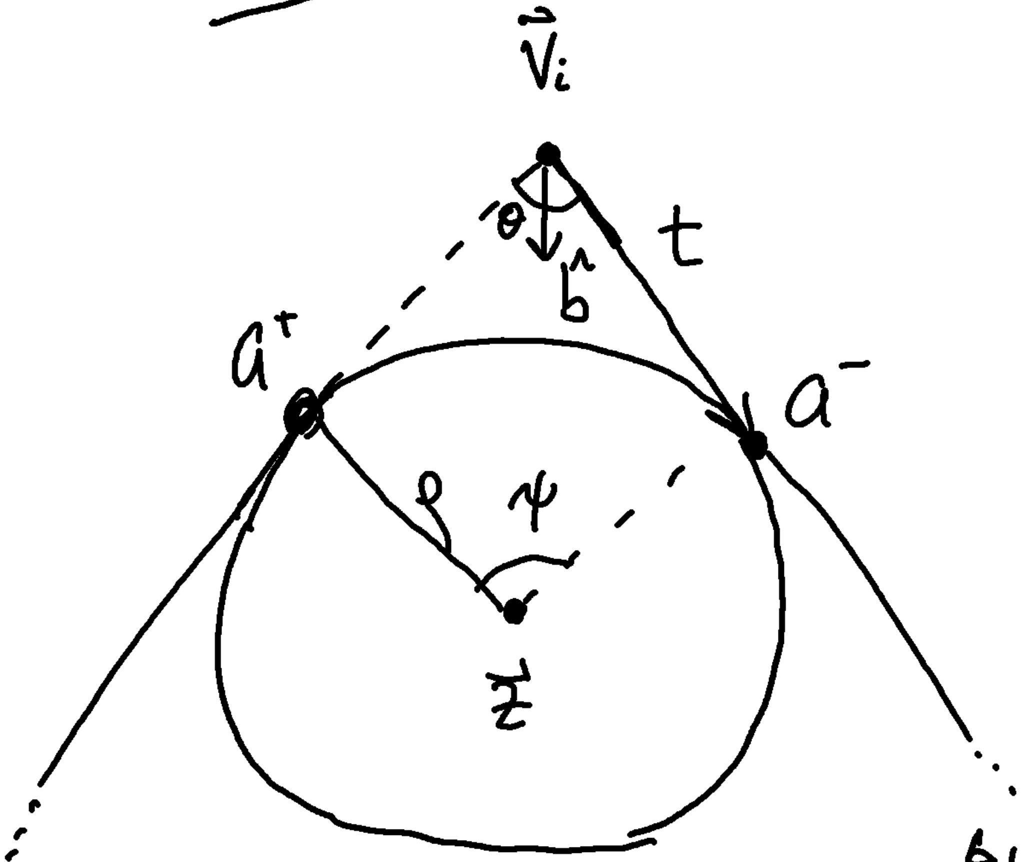
Rounded Definitions



$$\tan \theta/2 = \frac{\rho}{t}$$

$$\boxed{t = \rho / \tan \theta/2}$$

$$\sin \theta/2 = \frac{\rho}{|\vec{V}_i - \vec{z}|}$$

so

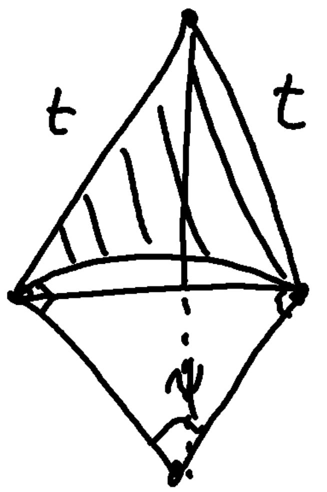
$$\boxed{\vec{z} = \vec{V}_i + \left(\frac{\rho}{\sin \theta/2}\right)\hat{b}.}$$

$$\boxed{\begin{aligned} \vec{a}^- &= \vec{V}_i - t\,\hat{e}_{z'(i)} \\ \vec{a}^+ &= \vec{V}_i + t\,\hat{e}_i \end{aligned}}$$

$$\boxed{\theta = \arccos\left(-\hat{e}_{z'(i)} \cdot \hat{e}_i\right)}$$

$$\boxed{\hat{b} = \frac{\hat{e}_i - \hat{e}_{z'(i)}}{|\hat{e}_i - \hat{e}_{z'(i)}|}}$$

$$\boxed{\psi = \pi - \theta.}$$



$$\boxed{\Delta\rho = -2t + \rho\psi.}$$

Find the area of the backlobe now with a line from a to a' and add only the circular segment.

$$\Delta A = \frac{1}{2}\psi\rho^2 - \frac{1}{2}\rho^2 \sin\psi$$

$$\boxed{\Delta A = \frac{1}{2}\rho^2(\psi - \sin\psi).}$$

Now the gradient

$$\frac{\partial \theta_k}{\partial V_j^\alpha} = \frac{\partial}{\partial V_j^\alpha} \arccos\left[-\hat{e}_{z'(k)} \cdot \hat{e}_k \right]$$

$$= \frac{(\delta_{kj} - \delta_{z'(k)j}) \left(\frac{\hat{e}_k^\alpha - (\hat{e}_k \cdot \hat{e}_{z'(k)}) e^\alpha_{z'(k)}}{|\vec{e}_{z'(k)}|} \right)}{\sin\theta_k}$$

$$+ \frac{(\delta_{z'(k)j} - \delta_{kj}) \left(\frac{\hat{e}^\alpha_{z'(k)} - (\hat{e}_k \cdot \hat{e}_{z'(k)}) e_k^\alpha}{|e_k|} \right)}{\sin\theta_k}$$

$$\boxed{\begin{aligned} \frac{\partial \theta_k}{\partial V_j^\alpha} &= \frac{(\delta_{kj} - \delta_{z'(k)j})}{\sin\theta_k} \left(\frac{\hat{e}_k^\alpha - \cos\theta_k \, \hat{e}^\alpha_{z'(k)}}{|\vec{e}_{z'(k)}|} \right) \\ &+ \frac{(\delta_{z(k)} - \delta_{kj})}{\sin\theta_k} \left(\frac{\hat{e}^\alpha_{z'(k)} - \cos\theta_k \, \hat{e}_k^\alpha}{|\vec{e}_k|} \right) \end{aligned}}$$

$$P = P_{\text{backbone}} - \sum_{j=1}^{n} \left(2t_j - \rho\psi_j\right)$$

$$\frac{\partial P}{\partial V_k^\alpha} = \sum_{j=1}^{n} \frac{\partial}{\partial V_k^\alpha}\left[|\vec{V}_{z(j)} - \vec{V}_j| - 2t_j + \rho\psi_j \right]$$

$$= \sum_{j=1}^{n}\left[\hat{e}_j^\alpha \left(\delta_{z(j)k} - \delta_{jk}\right) - 2\frac{\partial}{\partial V_k^\alpha}\frac{\rho}{\tan\theta_j/2} - \rho\frac{\partial\theta_j}{\partial V_k^\alpha}\right]$$

$$\frac{\partial P}{\partial V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + \rho\sum_{j=1}^{n}\frac{\partial\theta_j}{\partial V_k^\alpha}\left(-1 + \csc^2\theta_j/2\right)$$

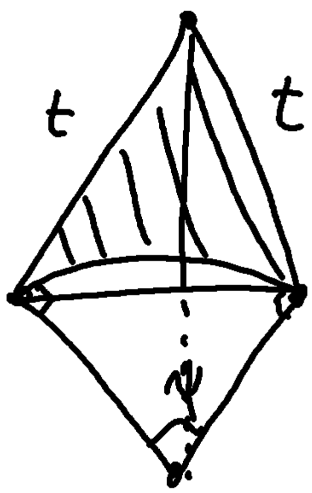
$$\boxed{\frac{\partial P}{\partial V_k^\alpha} = \hat{e}_{z'(k)}^\alpha - \hat{e}_k^\alpha + \rho\sum_{i=1}^{n}\frac{\partial\theta_i}{\partial V_k^\alpha}\cot^2\theta_i/2 \; .}$$

Next we separate this into a kite, wedge, and polygon

convexity

$$A = A_{backbone} - \sum_{j=1}^{n} \left(A_{kite,j} - A_{wedge}\right) S_j$$

$$= \sum_{j=1}^{n} \left[\frac{1}{2} \epsilon^{\alpha}{}_{\beta}\, e_j^{\beta}\, e^{\alpha}_{z(j)} - A_{kite,j} + \frac{\psi_j}{2}\rho^2 \right]$$



$$A_{kite} = \frac{1}{2}\left(2t\sin\theta/2\right)\left(\rho\cos\psi/2 + t\cos\theta/2\right)$$

$$= t\,\cancel{\sin\theta/2}\left(\rho\,\frac{\sin^2\theta/2 + \cos^2\theta/2}{\cancel{\sin\theta/2}}\right)$$

$$A_{kite} = t\rho$$

$$\boxed{A_{kite} = \frac{\rho^2}{\tan\theta/2}}$$

$$\boxed{A_{kite} - A_{wedge} = \rho^2\left(\cot\theta/2 - \psi/2\right)}$$

$$\boxed{\frac{\partial A}{\partial V_k{}^{\alpha}} = \frac{1}{2}\epsilon^{\alpha}{}_{\beta}\left(\vec{V}_{z(k)} - \vec{V}_{z'(k)}\right)^{\beta} + \frac{1}{2}\rho^2 \sum_{i=1}^{n} S_i \cot^2\left(\frac{\theta_i}{2}\right)\frac{\partial \theta_i}{\partial V_k{}^{\alpha}}.}$$

from $1-\csc^2$